

Fall Term 2024

Title:	Assignment 1	Due Date:	29-10-2024 23:59:59
Instructor:	Muhammad Yasin Nasir	Assig Date:	20-10-2024
Book:	Assembly Language Programming Organization of the IBM PC by Ytha Yu		
Subject:	Computer Organization & Assembly Language		
Syllabus:	Selected topics from Chapter 01,03,04,05,06		
Name:		Roll No.	

Instructions:

01. Questions are SELF EXPLANATORY. Understanding the Question is part of Solution.
02. Error in Question will be advantageous to Student.
03. Read the Questions carefully before attempting.
04. Attempt All Questions in a Precise Fashion.
05. Manage Your Time. Therefore, there will be no extension in the deadline.
06. Plagiarism may lead to F grade, so be honest.

Submission Method:

01. Assignment must be submitted individually.
02. **Take print out of the given assignment. Use A4 size paper** (210 × 297 millimeters or 8.27 × 11.69 inches) to print it out. Solve the assignment on the printed sheets.
03. Use any scanner tool to convert it to pdf. Text must have clear visibility after scanning. Blurred pdfs will be rejected.
04. Attempt the questions on the given space.
05. Upload the solved assignment (soft copy) at [university portal](#) before the deadline.
06. Submit the assignment (hard from) during the first half hour of the subsequent lecture.

Guidelines:

01. Observe the [ASCII table](#) . You have to follow character column.
02. Read the following article: [Fun with modular arithmetic](#). Modular concepts are used to formulate your questions.
03. Read the given example and understand the variables in the description.

First Name as String1							Middle Name as String 2					Last Name as String 3					
M	u	h	a	m	m	a	d	Y	a	s	i	n	N	a	s	i	r
↕							↕	↕				↕	↕				↕
0	1	2	3	4	5	6	7	0	1	2	3	4	0	1	2	3	4

Observe the above three strings. s_1 as Muhammad, s_2 as Yasin and, s_3 as Nasir are three strings.

- Each string is starting with 0 index. e.g. First index is 0 and last index is 7 for string 1.
- Each string must start with CAPITAL letter followed by small letters.

- Length of string is number of characters in the string, denoted by $Length(String)$. For the given strings, length of three strings are 8, 5 and 5 respectively.

04. Selection of string for the assignment.

- Use your registered name in the CS department (Used at SAP)
- Select first three strings of your name. e.g. Syed Abdul Wajid will be selected from Syed Abdul Wajid Bin Arish.
- If number of strings are lesser than three in your name, then
 - Use your first name as the second string.
 - Use your first name as the third string.

05. Evaluate the following values for your assignment to start proceedings.

String 1 = _____

String 2 = _____

String 3 = _____

06. Evaluate the following values for your assignment to start proceedings. Use characters in strings to evaluate each variable. Fill the values in the given table as per your requirement. Note: s_0 denotes the character at '0' index for string1, s_{11} denotes the character at '1' index for string1 & so on.

Variables	Values	Variables	Values	Variables	Values in Hexadecimal
$[ASCII(s_0)]_{16}$ $= x_0$		$x_0 \bmod 2$ $= y_0$		$x_0 \bmod (16)_{10}$ $= z_0$	
$[ASCII(s_1)]_{16}$ $= x_1$		$x_1 \bmod 2$ $= y_1$		$x_1 \bmod (16)_{10}$ $= z_1$	
$[ASCII(s_2)]_{16}$ $= x_2$		$x_2 \bmod 2$ $= y_2$		$x_2 \bmod 16$ $= z_2$	
$[ASCII(s_3)]_{16}$ $= x_3$		$x_3 \bmod 2$ $= y_3$		$x_3 \bmod 16$ $= z_3$	
$[ASCII(s_4)]_{16}$ $= x_4$		$x_4 \bmod 2$ $= y_4$		$x_4 \bmod 16$ $= z_4$	
$[ASCII(s_5)]_{16}$ $= x_5$		$x_5 \bmod 2$ $= y_5$		$x_5 \bmod 16$ $= z_5$	
$[ASCII(s_6)]_{16}$ $= x_6$		$x_6 \bmod 2$ $= y_6$		$x_6 \bmod 16$ $= z_6$	
$[ASCII(s_7)]_{16}$ $= x_7$		$x_7 \bmod 2$ $= y_7$		$x_7 \bmod 16$ $= z_7$	
$[ASCII(s_8)]_{16}$ $= x_8$		$x_8 \bmod 2$ $= y_8$		$x_8 \bmod 16$ $= z_8$	
$[ASCII(s_9)]_{16}$ $= x_9$		$x_9 \bmod 2$ $= y_9$		$x_9 \bmod 16$ $= z_9$	
$[ASCII(s_{1A})]_{16}$ $= x_{10}$		$x_{10} \bmod 2$ $= y_{10}$		$x_{10} \bmod 16$ $= z_{10}$	
$[ASCII(s_{1B})]_{16}$ $= x_{11}$		$x_{11} \bmod 2$ $= y_{11}$		$x_{11} \bmod 16$ $= z_{11}$	
$[ASCII(s_{1C})]_{16}$ $= x_{12}$		$x_{12} \bmod 2$ $= y_{12}$		$x_{12} \bmod 16$ $= z_{12}$	

Assuming the last variable in the above table using the characters in string1 is i .

Variables	Values	Variables	Values	Variables	Values in Hexadecimal
$[ASCII(s20)]_{16}$ $= x_{i+1}$		$x_{i+1} \bmod 2$ $= y_{i+1}$		$x_{i+1} \bmod 16$ $= z_{i+1}$	
$[ASCII(s21)]_{16}$ $= x_{i+2}$		$x_{i+2} \bmod 2$ $= y_{i+2}$		$x_{i+2} \bmod 16$ $= z_{i+2}$	
$[ASCII(s22)]_{16}$ $= x_{i+3}$		$x_{i+3} \bmod 2$ $= y_{i+3}$		$x_{i+3} \bmod 16$ $= z_{i+3}$	
$[ASCII(s23)]_{16}$ $= x_{i+4}$		$x_{i+4} \bmod 2$ $= y_{i+4}$		$x_{i+4} \bmod 16$ $= z_{i+4}$	
$[ASCII(s24)]_{16}$ $= x_{i+5}$		$x_{i+5} \bmod 2$ $= y_{i+5}$		$x_{i+5} \bmod 16$ $= z_{i+5}$	

Assuming the last variable using characters in string2 is j .

Variables	Values	Variables	Values	Variables	Values in Hexadecimal
$[ASCII(s30)]_{16}$ $= x_{j+1}$		$x_{j+1} \bmod 2$ $= y_{j+1}$		$x_{j+1} \bmod 16$ $= z_{j+1}$	
$[ASCII(s31)]_{16}$ $= x_{j+2}$		$x_{j+2} \bmod 2$ $= y_{j+2}$		$x_{j+2} \bmod 16$ $= z_{j+2}$	
$[ASCII(s32)]_{16}$ $= x_{j+3}$		$x_{j+3} \bmod 2$ $= y_{j+3}$		$x_{j+3} \bmod 16$ $= z_{j+3}$	
$[ASCII(s33)]_{16}$ $= x_{j+4}$		$x_{j+4} \bmod 2$ $= y_{j+4}$		$x_{j+4} \bmod 16$ $= z_{j+4}$	
$[ASCII(s34)]_{16}$ $= x_{j+5}$		$x_{j+5} \bmod 2$ $= y_{j+5}$		$x_{j+5} \bmod 16$ $= z_{j+5}$	

Q 1: For the 8086 architecture of general computer. Use the following address and it corresponding data table.

Address	Content Equation	Data
01F0	$y_1y_2y_3y_{i+1}y_{i+2}y_{i+3}y_{j+1}y_{j+2}$	
01F1	$y_1y_2y_5y_{j+1}y_{j+2}y_{j+3}y_{i+1}y_{i+2}$	
01F2	$y_{i+1}y_{i+2}y_{i+3}y_1y_2y_3y_{j+3}y_{j+2}$	
01F3	$y_{j+1}y_{j+2}y_{j+3}y_1y_2y_3y_{i+2}y_{i+2}$	
01F4	z_1z_{i+1}	
01F5	$z_{j+1}z_{j+2}$	
01F6	$z_{j+2}z_{i+2}$	
01F7	z_1z_{j+1}	
01F8	$z_{i+2}z_{j+3}$	
01F9	$z_{j+2}z_1$	
01FA	$z_{i+1}z_1$	
01FB	$z_{j+3}z_{j+1}$	

Part a:

i) What are the contents of memory word at address 01F2?

ii) What are the contents of memory word at address 01F8?

iii) What are the contents of memory word whose high byte is the byte at address 01F4?

iv) What are the contents of memory word whose high byte is the byte at address 01F2?

v) What are the contents of memory quad word whose high byte is the byte at address 01F6?

vi) What are the contents of memory double word whose low byte is the byte at address 01FA?

vii) What are the contents of memory quad word whose low byte is the byte at address 01F6?

viii) What are the contents of memory double word whose byte-2 is the byte at address 01F7?

ix) What are the contents of memory quad word whose high byte-3 is the byte at address 01F9?

x) What are the contents of memory quad word whose high byte-4 is the byte at address 01F3?

Part b: What is

i) Bit-7 of byte 01F2 ?

ii) Bit-0 of word 01F3 ?

iii) Bit-7 of byte 01F2 ?

iv) Bit-11 of word 01F2 ?

v) Low nibble of byte 01F2 ?

vi) High nibble of byte 01F3 ?

vii) Bit-20 of double word at 01F6 ?

viii) Bit-31 of quad word 01F6?

ix) High nibble of ten byte at 01F2 ?

x) Bit-41 of quad word 01F6 ?

Q 2: Assuming $offset = z_{j+1}z_{j+2}z_{i+2}z_{i+3}$ and $physical\ address = z_{i+2}z_{j+2}z_{i+2}z_{i+1}z_3$, then calculate the *segment address*.

Q 3: Assuming $segment\ address = z_i z_j z_{i-2} z_{i-3}$, and $offset\ address = z_2 z_{j+2} z_{i+2} z_3$, then calculate *physical address*.

Q 4: Assuming $segment\ address = z_{i-2} z_{j-2} z_{i+2} z_{i-1}$, and $physical\ address = z_{i+2} z_{j+2} z_{i+2} z_{i+1} z_3$, then calculate *offset address*.

Q 5: Assuming $AX = z_{i-2}z_{j-2}z_{i+2}z_{i-1}$ and $BX = z_i z_j z_{i-2} z_{i-3}$, perform the following operations in the given table and write down the status of flag bits as 1/0.

Note: Fill the given table assuming the status of register/ flag bits after execution.

	Default Flag bits		SF=0	ZF=0	CF=0	AF=0	OF=0	PF=0
	AX	BX	SF	ZF	CF	AF	OF	PF
AND AX, BX								
ADD AX, BX								
OR AX, BX								
SUB AX, BX								
AND AH, BL								
XOR BH, BH								
XOR AL, AL								
OR AH, AH								

Q 6: Assuming $AX = z_{j+1}z_{j+2}z_{i+2}z_{i+3}$ and $BX = z_{j+2}z_{i+2}z_{i+1}z_3$, perform the following operations in the given table and write down the status of flag bits as 1/0.

Note: Fill the given table assuming the status of register/ flag bits after execution.

	Default Flag bits		SF=1	ZF=1	CF=1	AF=1	OF=1	PF=1
	AX	BX	SF	ZF	CF	AF	OF	PF
AND AX, BX								
ADD AX, BX								
OR AX, BX								
SUB AX, BX								
AND AH, BL								
XOR BH, BH								
XOR AL, AL								
OR AH, AH								